

The relativistik kinetic energy is

$$E_K = \sqrt{(pc)^2 + (mc^2)^2} - mc^2.$$

If we rewrite this to a hamiltonian using $p = i\hbar\nabla$ we get

$$\begin{aligned} H &= \sqrt{(i\hbar c\nabla)^2 + (mc^2)^2} - mc^2. \\ &= \sqrt{(mc^2)^2 - (\hbar c)^2 \nabla^2} - mc^2 \end{aligned}$$

Shrödinger becomes:

$$i\hbar \frac{\partial}{\partial t} \psi = \sqrt{(mc^2)^2 - (\hbar c)^2 \nabla^2} \psi - mc^2 + V\psi$$

Here we have the sqrt of of the gradient squared, which is hard to calculate. To do this we will use fourier transformation to go from position space to momentum space. This will transform the space and momentum operators to (Griffiths 2018, 157)[1]

$$\begin{aligned} \phi(p) &= \langle p | S(t) \rangle = \int dx \langle p | x \rangle \langle x | S(t) \rangle = \int dx \langle p | x \rangle \psi(x), \\ \langle p | x \rangle &= \langle x | p \rangle^* = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}. \end{aligned}$$

But in momentum space the position and momentum operators are different (Griffiths 2018, 151)[1]

$$\begin{aligned} \hat{x} &= i\hbar \frac{\partial}{\partial p}, \\ \hat{p} &= p. \end{aligned}$$

To find the hamiltonian in momentum space we use

$$\langle p | \hat{H} | \phi \rangle = \int dp' \langle p | \hat{H} | p' \rangle \langle p' | \phi \rangle,$$

since the kinetic energy is only dependant on the momentum we get

$$\begin{aligned} \langle p | f(p) | p' \rangle &= f(p) \delta(p - p'), \\ \langle p | \hat{K} | \phi \rangle &= \hat{K} \phi(p). \end{aligned}$$

but the Coulumb potential must be rewritten. We do this by using the Yukawa trick. [2]

$$\begin{aligned} \langle g' | f(|w^T \mathbf{r}|) | g \rangle &= \langle g' | g \rangle \left(\frac{\beta}{\pi} \right)^{\frac{3}{2}} 2\pi \frac{e^{-\beta q^2}}{\beta q} \int_0^\infty r dr f(r) e^{-\beta r^2} \sinh(2\beta q r), \\ &= \langle g' | g \rangle J[f], \end{aligned}$$

which for Coulumb potential can be expressed as

$$= \langle g' | g \rangle J \left[\frac{1}{r} \right] = \langle g' | g \rangle \frac{erf(\sqrt{\beta} q)}{q} \xrightarrow{q \rightarrow 0} \langle g' | g \rangle \frac{2}{\sqrt{\pi}} \sqrt{\beta}.$$

This gives us the final potential operator

$$\langle p' | \hat{V}(\mathbf{r}) | p \rangle = \langle p' | p \rangle \frac{2}{\sqrt{\pi}} \sqrt{\beta}$$

where

$$\beta = (w^T B^{-1} w)^{-1}$$

Use the variation principle to "guess" a ground state wave function, where we use the Jacobian to reduce 3 body problem 2 two body problem. We use a general gaussian wavefunction

$$\psi(\alpha, \beta, x, y) = e^{-\alpha x^2 - \beta y^2},$$

where x is the distance from pion to center of nucleon and x is distance between proton and neutron (I think).

To improve the "trial function" we use a linear combination of these gaussians

Litteratur

- [1] Griffiths, D. J. 2018, "Introduction to Quantum Mechanics"
- [2] Dimitri, F. "Analytic matrix elements with shifted correlated Gaussians"